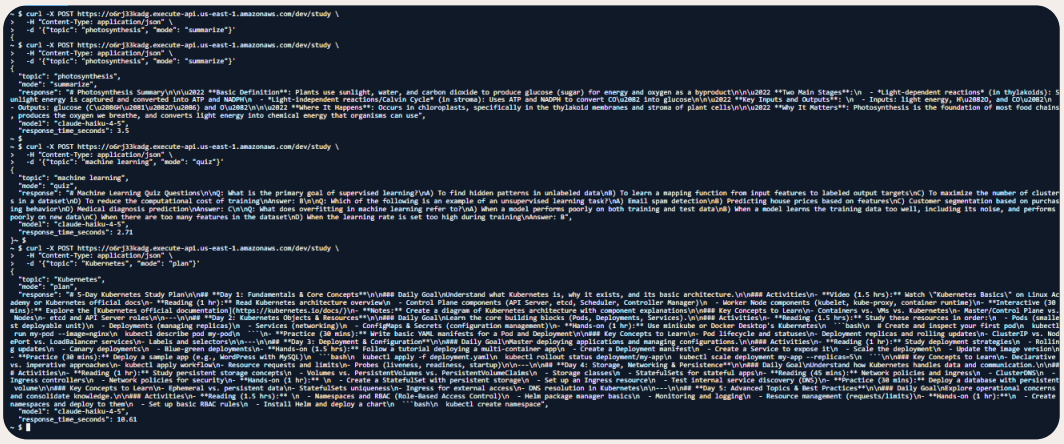
BUILD A STUDY BUDDY API WITH LAMBDA

By Nikhila Reddy | October 2026



Project Vision: Building a Serverless AI Study API

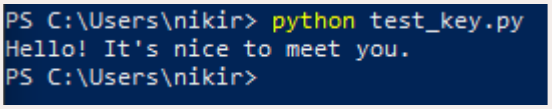
Goals and motivation

In this project, I'm building a serverless Study Buddy REST API using AWS Lambda, API Gateway, and an AI model so that I can generate study materials such as summaries, quizzes, and study plans through an HTTPS endpoint.

Connecting to Claude: API Key Setup and First AI Response

Setting up the Anthropic API key

In this step, I'm setting up an Anthropic API account and generating an API key so that I can connect my Study Buddy API to Claude and generate AI-powered study materials.



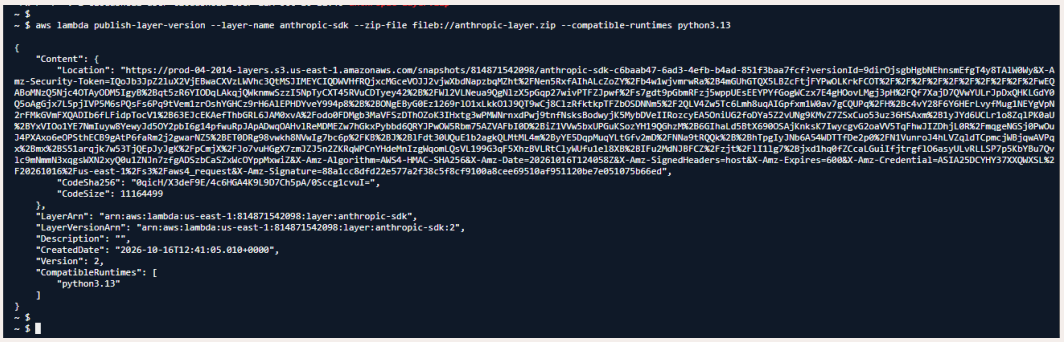
Test script results and model used

When I ran test_key.py, I saw a successful response from Claude confirming that my API key worked, using the model claude-haiku-4-5.

Packaging Dependencies: Building the Lambda Layer in CloudShell

Why a Lambda layer is needed

In this step, I'm building a Lambda Layer containing the Anthropic Python SDK so that my Lambda function can import the required library and communicate with Claude's API.



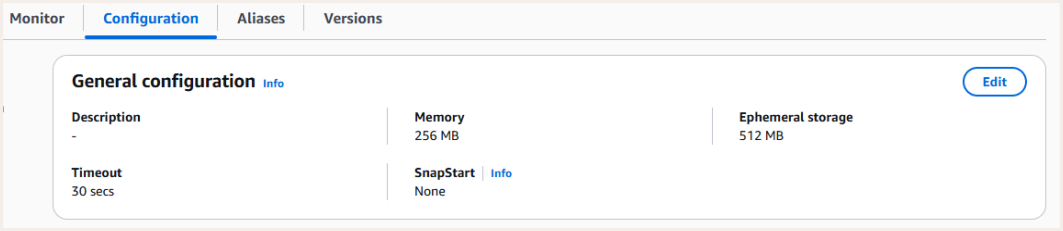
The python/ directory structure explained

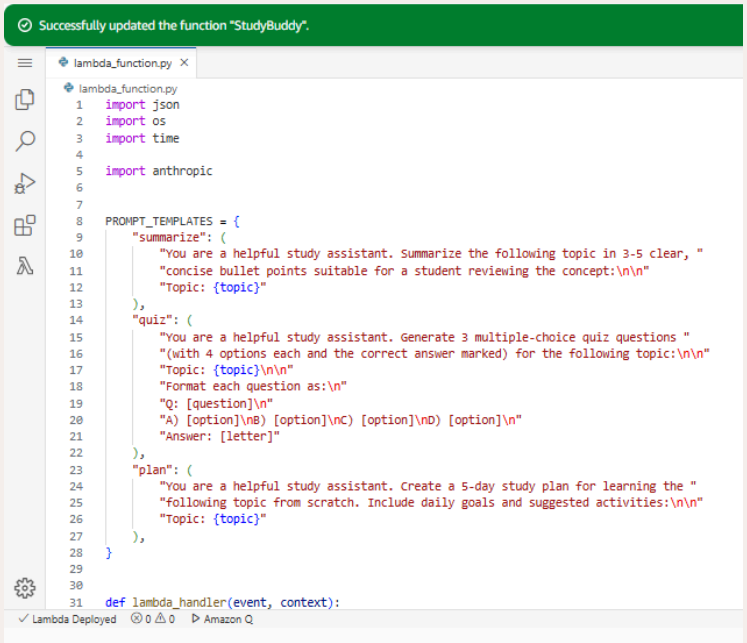
The python/ directory is required because AWS Lambda expects Python libraries in a specific folder structure when using a Lambda layer. By placing the Anthropic Python SDK and its dependencies inside the top-level python/ directory, Lambda can locate and load them when the function runs. This allows our Lambda function to import the SDK and use it to communicate with Claude's API without packaging all the libraries directly with the function code.

Deploying the Lambda Function with AI Prompt Routing

Function setup and configuration

In this step, I'm creating an AWS Lambda function with Python 3.14, attaching the Anthropic SDK layer, configuring the API key and function settings, and adding the handler code so that my function can process study requests and use Claude to generate summaries, quizzes, and study plans.

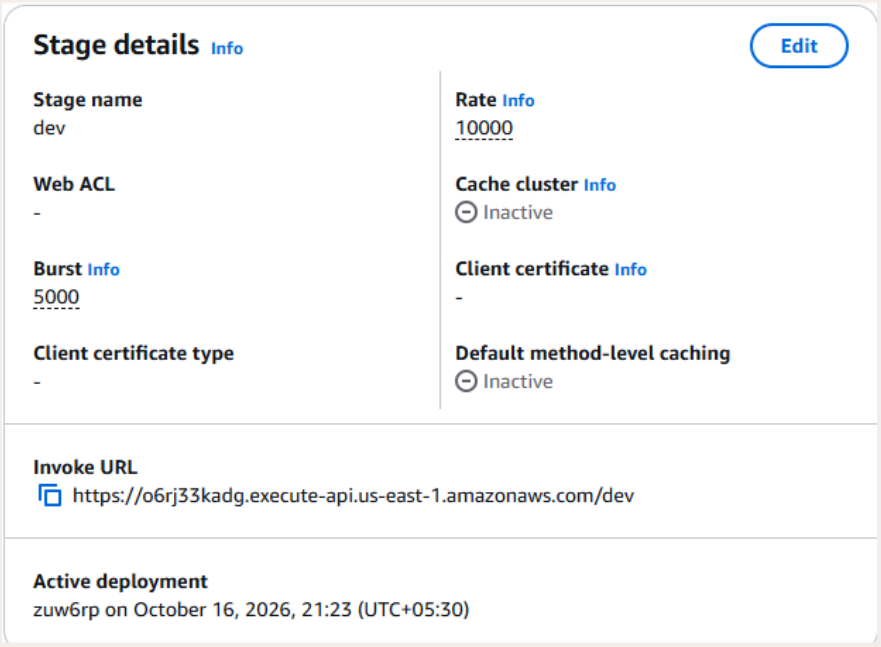




Exposing the API: API Gateway Configuration and Deployment

Setting up REST API with Lambda proxy integration

In this step, I'm setting up an API Gateway REST API with a /study resource and a POST method connected to my StudyBuddy Lambda function, so that I can receive study requests through an HTTPS endpoint and pass them to Lambda for processing.



Live invoke URL

My full endpoint URL is https://https://o6rj33kadg.execute-api.us-east-1.amazonaws.com/dev/study

Testing the Study Buddy: Live AI Responses Across All Modes

Testing strategy and goals

In this step, I'm testing my Study Buddy API with different study modes and checking its responses, so that I can confirm the API processes requests correctly and returns the expected study materials.

Error handling validation

When I send a request without a topic, the API returns a 400 Bad Request error with the message "Missing required field: topic" and a usage hint showing the correct JSON format.

Bonus: Request Logging with DynamoDB and a /history Endpoint



Why request logging matters in production

In this project extension, I learned that logging requests helps with monitoring API usage, troubleshooting errors, investigating issues, and understanding performance. By recording details such as the topic, request mode, timestamp, and response time, developers can review recent requests and maintain the API more reliably.

Reflections and Key Takeaways

Tools and concepts mastered

The key tools I used include AWS Lambda, API Gateway, DynamoDB, AWS CloudShell, IAM, and the Anthropic API. Key concepts I learnt include building a serverless REST API, connecting Lambda to an AI model, using Lambda Layers to manage Python dependencies, handling API requests and errors, and logging requests in DynamoDB to review API usage and performance.

Time and challenges

This project took me approximately 2.5 hours, including the Anthropic API setup, implementation, testing, and documentation. The most challenging part was configuring the AWS services and permissions correctly and troubleshooting the integration to ensure the API worked as expected.

Looking ahead

I did this project today to learn how to build a serverless AI-powered API using AWS Lambda, API Gateway, and the Anthropic API, and how to store and retrieve request history using DynamoDB. Another skill I want to learn is building AI agents and automation workflows that solve practical, everyday problems.